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10CFR50.73

GNRO-2019/00035

August 7, 2019

U.S. Nuclear Regulatory Commission  
Attn: Document Control Desk  
Washington, DC 20555-0001

SUBJECT: Supplemental Licensee Event Report 2018-001-01, Reactor Manual  
Shutdown due to Turbine Pressure Control Valve Position Changes  
Grand Gulf Nuclear Station, Unit 1  
Docket No. 50-416  
License No. NPF-29

Dear Sir or Madam:

Attached is Supplemental Licensee Event Report 2018-001-01, Reactor Manual Shutdown due to Turbine Pressure Control Valve Changes. This report is being submitted in accordance with 10CFR50.73(a)(2)(iv)(A) for any event or condition that resulted in a manual or automatic actuation of any of the systems listed in 10CFR50.73(a)(2)(iv)(B).

This letter contains no new commitments. If you have any questions or require additional information, please contact Jim Shaw at 601-437-2103.

Sincerely,

A handwritten signature in black ink, appearing to read "E. A. Larson".

Eric A. Larson  
EAL/rtw

Attachment: Supplemental Licensee Event Report 2018-001-01

cc:

NRC Region IV – Regional Administrator  
NRC Senior Resident Inspector, Grand Gulf Nuclear Station  
NRR Project Manager

Attachment

Supplemental Licensee Event Report 2018-001-01



## LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollects.Resource@nrc.gov](mailto:Infocollects.Resource@nrc.gov), and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

|                                                        |                              |                   |
|--------------------------------------------------------|------------------------------|-------------------|
| 1. Facility Name<br>Grand Gulf Nuclear Station, Unit 1 | 2. Docket Number<br>05000416 | 3. Page<br>1 OF 3 |
|--------------------------------------------------------|------------------------------|-------------------|

4. Title  
Reactor Manual Shutdown Due to Turbine Pressure Control Valve Position Changes

| 5. Event Date |     |      | 6. LER Number |                   |         | 7. Report Date |     |      | 8. Other Facilities Involved |               |
|---------------|-----|------|---------------|-------------------|---------|----------------|-----|------|------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Rev No. | Month          | Day | Year | Facility Name                | Docket Number |
| 01            | 30  | 2018 | 2018          | 001               | 01      | 08             | 07  | 2019 | N/A                          | 05000 N/A     |
|               |     |      |               |                   |         |                |     |      | Facility Name                | Docket Number |
|               |     |      |               |                   |         |                |     |      | N/A                          | 05000 N/A     |

|                                 |                                                                                               |                                             |                                                                                |                                               |  |  |  |  |
|---------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------|--|--|--|--|
| 9. Operating Mode<br><br>MODE 1 | 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply) |                                             |                                                                                |                                               |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2201(b)                                                           | <input type="checkbox"/> 20.2203(a)(3)(i)   | <input type="checkbox"/> 50.73(a)(2)(ii)(A)                                    | <input type="checkbox"/> 50.73(a)(2)(viii)(A) |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2201(d)                                                           | <input type="checkbox"/> 20.2203(a)(3)(ii)  | <input type="checkbox"/> 50.73(a)(2)(ii)(B)                                    | <input type="checkbox"/> 50.73(a)(2)(viii)(B) |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2203(a)(1)                                                        | <input type="checkbox"/> 20.2203(a)(4)      | <input type="checkbox"/> 50.73(a)(2)(iii)                                      | <input type="checkbox"/> 50.73(a)(2)(ix)(A)   |  |  |  |  |
| 10. Power Level<br><br>91%      | <input type="checkbox"/> 20.2203(a)(2)(i)                                                     | <input type="checkbox"/> 50.36(c)(1)(i)(A)  | <input checked="" type="checkbox"/> 50.73(a)(2)(iv)(A)                         | <input type="checkbox"/> 50.73(a)(2)(x)       |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2203(a)(2)(ii)                                                    | <input type="checkbox"/> 50.36(c)(1)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(v)(A)                                     | <input type="checkbox"/> 73.71(a)(4)          |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2203(a)(2)(iii)                                                   | <input type="checkbox"/> 50.36(c)(2)        | <input type="checkbox"/> 50.73(a)(2)(v)(B)                                     | <input type="checkbox"/> 73.71(a)(5)          |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2203(a)(2)(iv)                                                    | <input type="checkbox"/> 50.46(a)(3)(ii)    | <input type="checkbox"/> 50.73(a)(2)(v)(C)                                     | <input type="checkbox"/> 73.77(a)(1)          |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2203(a)(2)(v)                                                     | <input type="checkbox"/> 50.73(a)(2)(i)(A)  | <input type="checkbox"/> 50.73(a)(2)(v)(D)                                     | <input type="checkbox"/> 73.77(a)(2)(ii)      |  |  |  |  |
|                                 | <input type="checkbox"/> 20.2203(a)(2)(vi)                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(B)  | <input type="checkbox"/> 50.73(a)(2)(vii)                                      | <input type="checkbox"/> 73.77(a)(2)(iii)     |  |  |  |  |
|                                 |                                                                                               | <input type="checkbox"/> 50.73(a)(2)(i)(C)  | <input type="checkbox"/> Other (Specify in Abstract below or in NRC Form 366A) |                                               |  |  |  |  |

## 12. Licensee Contact for this LER

|                                                             |                                                        |
|-------------------------------------------------------------|--------------------------------------------------------|
| Licensee Contact<br>Jim Shaw, Manager, Regulatory Assurance | Telephone Number (Include Area Code)<br>(601) 437-2103 |
|-------------------------------------------------------------|--------------------------------------------------------|

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component                | Manufacturer | Reportable To ICES | Cause | System | Component | Manufacturer | Reportable To ICES |
|-------|--------|--------------------------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| B     | H13    | 1H13P825<br>JC05<br>C332 | Siemens      | Yes                | N/A   | N/A    | N/A       | N/A          | N/A                |

## 14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

## 15. Expected Submission Date

| Month | Day | Year |
|-------|-----|------|
| N/A   | N/A | N/A  |

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On January 30, 2018, Grand Gulf Nuclear Station (GGNS) was operating in Mode 1 at 91% power and was in the process of increasing power. At 1739 hours CST on January 30, 2018, GGNS experienced main turbine control valve (TCV) position changes of between 3 and 5 percent and main generator output load changes of approximately 20 megawatts with a periodicity of approximately 3 seconds.

At 1822 on January 30, 2018, operators performed a manual shut down of the reactor by moving the reactor mode switch from RUN to SHUTDOWN to place the unit in Mode 3. Upon reactor shutdown all systems performed as designed and no subsequent safety system actuations occurred. The TCV position changes were caused by an inappropriately high gain setting in one channel of the TCVs resonance compensator circuit. Corrective actions include adjustments to the control circuit, and revision of work task documents for adjustment of the control circuits.

There were no consequences to the general safety of the public, nuclear safety, industrial safety and radiological safety for this event.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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| 1. FACILITY NAME                   | 2. DOCKET NUMBER | 3. LER NUMBER |                   |         |
|------------------------------------|------------------|---------------|-------------------|---------|
|                                    |                  | YEAR          | SEQUENTIAL NUMBER | REV NO. |
| Grand Gulf Nuclear Station, Unit 1 | 05000-416        | 2018          | - 001             | - 001   |

**NARRATIVE****A. PLANT CONDITIONS PRIOR TO THE EVENT**

Grand Gulf Nuclear Station (GGNS) Unit 1 was operating at 91% power in Mode 1 and increasing power. There were no Structures, Systems, or Components that were inoperable that contributed to the event.

**B. DESCRIPTION**

At 1739 hours CST on January 30, 2018, while the plant was operating at approximately 91% reactor power/1322 megawatts electric power (mwe), operators observed a 20 mwe electrical oscillation (peak to peak) with a periodicity of approximately 3 seconds. Concurrent oscillations of between 3 and 5% were noted on all four turbine control valves.

At 1822 hours on January 30, 2018, operators performed a manual shutdown of the reactor by moving the reactor mode switch from RUN to SHUTDOWN. Upon reactor shutdown all systems performed as designed and no subsequent safety system actuations occurred.

**C. REPORTABILITY**

This report is made pursuant to 10CFR50.73(a)(2)(iv)(A) for any event or condition that resulted in a manual or automatic actuation of any of the systems listed in 10CFR50.73(a)(2)(iv)(B). This event was reported under 10CFR50.72(b)(3)(iv)(B)(1) in ENS notification 53115.

**D. CAUSE**

The cause of the reactor scram was a manual shutdown of the reactor due to operator concerns about main generator output variations, caused by unanticipated turbine pressure control valve (TCV) motion.

The turbine pressure control system controls turbine speed, operates the steam bypass system to keep reactor pressure within limits and avoid transients, and controls main turbine inlet pressure.

The unanticipated TCV motion was caused by an inappropriately high gain setting in the main steam line resonance compensator circuit. Investigation revealed that the high gain setting of the Resonance Compensator circuitry occurred because the work order steps used to calibrate the card when it was installed in June of 2016 were incorrect. The incorrect setting did not create an error of such magnitude as to be automatically excluded from the circuit internal diagnostics, but did create enough compensatory response from the TCV to be discernable by the operating crew.

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

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|------------------------------------|------------------|---------------|----------------------|------------|
| Grand Gulf Nuclear Station, Unit 1 | 05000-416        | YEAR          | SEQUENTIAL<br>NUMBER | REV<br>NO. |
|                                    |                  | 2018          | - 001                | - 001      |

**NARRATIVE****E. CORRECTIVE ACTIONS**

The following corrective actions are completed or planned.

Completed:

- The main steam line resonance compensator circuit was recalibrated.
- Work instructions to setup steam line compensator were revised to provide corrected work steps.

**F. SAFETY SIGNIFICANCE**

The safety significance of this event is low because all systems operated as designed and the TCV changes did not challenge any safety parameters. The manual scram was performed as a conservative measure. There were no actual nuclear safety consequences or radiological consequences during the event.

**G. PREVIOUS SIMILAR EVENTS**

Entergy conducted a three-year review of the relevant licensee event reports and determined that there was one similar known event reported as GGNS Licensee Event Report 2016-05-00 "Automatic Reactor SCRAM." The failure in that event was a failed amplifier card, which is different from this event which was an incorrect potentiometer setting. Therefore, the corrective actions for the previous event would not have prevented this event.